STAT 20000
Ch 17 The Expected Value and Standard Error
Ch 18 The Normal Approximation for Probability Histograms

17.1 The Expected Value for the Sum of the Draws

Suppose 40 draws are to be made (at random with replacement) from the box

$$\boxed{\$2}\ \boxed{-\$1}\ \boxed{-\$1}\ \boxed{-\$1}$$

You'd expect each tickets to be drawn about ______ times, so the sum of the 40 draws should be around

$$\$2\times\_\_\_\_+(-\$1)\times\_\_\_\_+(-\$1)\times\_\_\_\_+(-\$1)\times\_\_\_\_ = \_\_\_\_\_\_\_\_.$$

Another way to reach the same conclusion is to use the fact that the average of the numbers in the box is

$$\frac{\$2+(-\$1)+(-\$1)+(-\$1)}{4} = -\$0.25.$$

On the average, you lose \$0.25 on each draw.

In 40 draws you can expect to lose around $40 \times \$0.25 = \$10$.

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You'd expect each tickets to be drawn about __10__ times, so the sum of the 40 draws should be around

$$\$2 \times \underline{\ 10\ } + (-\$1) \times \underline{\ 10\ } + (-\$1) \times \underline{\ 10\ } + (-\$1) \times \underline{\ 10\ } = \underline{\ -\$10\ }.$$

Another way to reach the same conclusion is to use the fact that the average of the numbers in the box is

$$\frac{\$2 + (-\$1) + (-\$1) + (-\$1)}{4} = -\$0.25.$$

On the average, you lose \$0.25 on each draw.

In 40 draws you can expect to lose around $40 \times \$0.25 = \$10$.

The Expected Value for the Sum of the Draws (Cont'd)

Why do the two methods give the same conclusion?

$$
\begin{aligned}
&\$2 \times 10 + (-\$1) \times 10 + (-\$1) \times 10 + (-\$1) \times 10 && \text{(first way)} \\
&= 10 \times \Big(\$2 + (-\$1) + (-\$1) + (-\$1)\Big) \\
&= 40 \times \frac{1}{4} \times \Big(\$2 + (-\$1) + (-\$1) + (-\$1)\Big) \\
&= 40 \times \left(\frac{\$2 + (-\$1) + (-\$1) + (-\$1)}{4}\right) && \text{(second way)}
\end{aligned}
$$

In general, the **expected value** for the **sum of draws** made at random with replacement from a box is

$$(\text{number of draws}) \times (\text{average of box})$$

Here "average of box" is short for the average of the numbers in the box.

Example 1: Rolling a Pair of Dice

What is the expected number of the sum of spots when rolling a pair of dice?

- This is the same as the expected value for the sum of 2 draws from the box

$$\boxed{\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}\ \boxed{6}}$$

- The number of draws equals _____.
- The average of the box equals

$$\frac{1+2+3+4+5+6}{6} = 3.5.$$

- The answer is _____ $\times$ _______ $=$ _____.

Example 1: Rolling a Pair of Dice

What is the expected number of the sum of spots when rolling a pair of dice?

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- The number of draws equals $\underline{\quad 2 \quad}$.
- The average of the box equals

$$\frac{1+2+3+4+5+6}{6} = 3.5.$$

- The answer is $\underline{\qquad\quad} \times \underline{\qquad\qquad} = \underline{\qquad\quad}$.

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- The number of draws equals __2__.
- The average of the box equals

$$\frac{1+2+3+4+5+6}{6} = 3.5.$$

- The answer is __2__ $\times$ __3.5__ $=$ __7__.

Example: Roulette revisited

Suppose you are going to play roulette 10 times, staking \$1 on red-and-black each time. This bet pays 1 to 1, and you have 18 chances in 38 of winning. Your net gain is like the sum of ______ draws from the box

$$\boxed{18 \text{ tickets of } \boxed{\$1} \quad 20 \text{ tickets of } \boxed{-\$1}}$$

The expected value of your net gain is ___________

- The average of the box equals

$$\frac{18 \times (\$1) + 20 \times (-\$1)}{38} = -\$\frac{1}{19} \approx -\$0.0526$$

- The expected net gain is

$$\text{number of draws} \times (\text{Ave. of the box})$$
$$\approx \_\_\_\_\_ \times \_\_\_\_\_\_\_\_\_\_\_\_ \approx \_\_\_\_\_\_\_\_\_\_\_$$

Example: Roulette revisited

Suppose you are going to play roulette 10 times, staking \$1 on red-and-black each time. This bet pays 1 to 1, and you have 18 chances in 38 of winning. Your net gain is like the sum of <u>10</u> draws from the box

$$\boxed{\text{18 tickets of } \boxed{\$1} \quad \text{20 tickets of } \boxed{-\$1}}$$

The expected value of your net gain is <u>$-\$0.526$</u>

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$$\frac{18 \times (\$1) + 20 \times (-\$1)}{38} = -\$\frac{1}{19} \approx -\$0.0526$$

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$$\text{number of draws} \times (\text{Ave. of the box})$$

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18 tickets of $\boxed{\$1}$ 20 tickets of $\boxed{-\$1}$

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- The expected net gain is

$$\text{number of draws} \times (\text{Ave. of the box})$$
$$\approx \underline{10} \times \underline{(-\$0.0526)} \approx \underline{-\$0.526}$$

17.2 Standard Error for the Sum of the Draws

When 40 draws are to be made (at random w/ replacement)from the box

$$\boxed{\boxed{2}\ \boxed{-1}\ \boxed{-1}\ \boxed{-1}}$$

the observed sum of the draws will be off from its expected value ($-\$10$) by some **chance error**.

$$\text{(observed) sum of the draws} = \text{expected value} + \text{chance error.}$$

- The value of "chance error" may change from time to time
- The long-run "average" size of these chance errors is called the **standard error** for the sum of the draws.
  - "long-run" means to repeat the same process many times (One repetition = 40 draws from the box)
  - "average" actually refers to "root-mean-squares"
  - Say 1000 repetitions were made. One could get 1000 chance errors. The root-mean-square of these 1000 chance errors is roughly the **standard error**.

Standard Error for the Sum of the Draws (Cont'd)

The **standard error** for the sum of the draws turns out to be

$$\sqrt{\text{number of draws}} \times (\text{SD of the box})$$

Here "SD of box" is short for **the SD of the numbers in the box**. Why?

In particular, the formula says that the likely size of the chance error for a single draw is just the SD of the box. Here's the intuition.

- On a single draw, you'd expect to get the average of the box.
- So the chance error in a single draw is the difference between the number drawn and the average of the box.
- The likely size of such a chance error is the amount by which the tickets in the box typically deviate from their average.
- That's just the SD of the box.

Standard Error for the Sum of the Draws (Cont'd)

- I just argued that the variability in a single draw is the same as the variability of the tickets in the box. It's intuitively clear that the sum of n draws should be more variable than a single draw. Why is there only $\sqrt{n}$ times as much variability, and not n times as much variability?

- Intuition can only provide a qualitative answer. On each draw there is an individual chance error, equal to the number drawn minus the average of the box. The chance error in the sum of the draws equals the sum of these individual chance errors. The point is that some of the individual errors are positive, and some negative. By and large, the positive individual errors cancel out the negative individual errors, so that the chance error in the sum grows slowly with respect to n.

Example

Suppose 40 draws are to be made (at random with replacement) from the box

$$\boxed{2}\ \boxed{-1}\ \boxed{-1}\ \boxed{-1}$$

You'd expect the sum of the draws to be about _________, **give or take** _________.

- SD of box:
  - numbers in the box: 2, -1, -1, -1
  - average of the numbers $= (2-1-1-1)/4 = -0.25$
  - deviations from the average: 2.25, -0.75, -0.75, -0.75
  - SD of box =

$$\sqrt{\frac{(2.25)^2 + (-0.75)^2 + (-0.75)^2 + (-0.75)^2}{4}} \approx 1.30$$

- Standard error for the sum =

$$\sqrt{\text{number of draws}} \times (\text{SD of the box}) = \sqrt{} \times \qquad =$$

Example

Suppose 40 draws are to be made (at random with replacement) from the box

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You'd expect the sum of the draws to be about $\underline{\quad -\$10 \quad}$, **give or take** $\underline{\qquad\qquad}$.

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  - numbers in the box: $2, -1, -1, -1$
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  - deviations from the average: $2.25, -0.75, -0.75, -0.75$
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$$\sqrt{\frac{(2.25)^2 + (-0.75)^2 + (-0.75)^2 + (-0.75)^2}{4}} \approx 1.30$$

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$$\sqrt{\frac{(2.25)^2 + (-0.75)^2 + (-0.75)^2 + (-0.75)^2}{4}} \approx 1.30$$

- Standard error for the sum $=$

$$\sqrt{\text{number of draws}} \times (\text{SD of the box}) = \sqrt{40} \times 1.30 = 8.22.$$

Example

Suppose 40 draws are to be made (at random with replacement) from the box

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- Standard error for the sum $=$

$$\sqrt{\text{number of draws}} \times (\text{SD of the box}) = \sqrt{40} \times 1.30 = 8.22.$$

Shortcut Formula for Box w/ Only 2 Kinds of Tickets

When the tickets in a box show **only two** different numbers, the SD of the box works out to

$$\left(\begin{matrix}\text{bigger}\\\text{number}\end{matrix} - \begin{matrix}\text{smaller}\\\text{number}\end{matrix}\right) \times \sqrt{\left(\begin{matrix}\text{fraction with}\\\text{bigger number}\end{matrix}\right) \times \left(\begin{matrix}\text{fraction with}\\\text{smaller number}\end{matrix}\right)}.$$

For example, consider the box $\boxed{\boxed{2}\ \boxed{-1}\ \boxed{-1}\ \boxed{-1}}$

- SD of box ≈ 1.30 (from the previous page)
- The short cut can be used since the tickets show only two different numbers, 2 and -1.
  - bigger number = ____, fraction = ______ .
  - smaller number = _____, fraction = ______.
- The SD of the box equals

$$(2-(-1)) \times \sqrt{\frac{1}{4} \times \frac{3}{4}} \approx 1.30 \ ,$$

in agreement with our previous calculation.

Shortcut Formula for Box w/ Only 2 Kinds of Tickets

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For example, consider the box $\boxed{\boxed{2}\ \boxed{-1}\ \boxed{-1}\ \boxed{-1}}$

- SD of box ≈ 1.30 (from the previous page)
- The short cut can be used since the tickets show only two different numbers, 2 and -1.
  - bigger number = $\underline{\ 2\ }$, fraction = $\underline{\ 1/4\ }$.
  - smaller number = $\underline{\ -1\ }$, fraction = $\underline{\ 3/4\ }$.
- The SD of the box equals

$$(2-(-1))\times\sqrt{\frac{1}{4}\times\frac{3}{4}}\approx 1.30\ ,$$

in agreement with our previous calculation.

Exercises

Find the SD for each of the three boxes below.

- $\boxed{\boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{1}\ \boxed{1}}$

- $\boxed{\boxed{-1}\ \boxed{1}\ \boxed{1}\ \boxed{1}\ \boxed{1}}$

- $\boxed{\text{1 ticket } \boxed{\$180} \quad \text{215 tickets } \boxed{-\$1}}$

Exercises

Find the SD for each of the three boxes below.

- $\boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{1}\ \boxed{1}$

$$(1-0)\times\sqrt{\frac{2}{5}\times\frac{3}{5}}=\frac{\sqrt{6}}{5}\approx 0.49.$$

- $\boxed{-1}\ \boxed{1}\ \boxed{1}\ \boxed{1}\ \boxed{1}$

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Exercises

Find the SD for each of the three boxes below.

- $\boxed{\boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{1}\ \boxed{1}}$

$$(1-0) \times \sqrt{\frac{2}{5} \times \frac{3}{5}} = \frac{\sqrt{6}}{5} \approx 0.49.$$

- $\boxed{\boxed{-1}\ \boxed{1}\ \boxed{1}\ \boxed{1}\ \boxed{1}}$

$$(1-(-1)) \times \sqrt{\frac{1}{5} \times \frac{4}{5}} = \frac{4}{5} = 0.8.$$

- $\boxed{\text{1 ticket } \boxed{\$180} \quad \text{215 tickets } \boxed{-\$1}}$

Exercises

Find the SD for each of the three boxes below.

- $\boxed{\boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{1}\ \boxed{1}}$

$$(1-0)\times\sqrt{\frac{2}{5}\times\frac{3}{5}}=\frac{\sqrt{6}}{5}\approx 0.49.$$

- $\boxed{\boxed{-1}\ \boxed{1}\ \boxed{1}\ \boxed{1}\ \boxed{1}}$

$$(1-(-1))\times\sqrt{\frac{1}{5}\times\frac{4}{5}}=\frac{4}{5}=0.8.$$

- $\boxed{\text{1 ticket } \boxed{\$180} \quad \text{215 tickets } \boxed{-\$1}}$

$$(\$180-(-\$1))\times\sqrt{\frac{215}{216}\times\frac{1}{216}}=\frac{\$181\sqrt{215}}{216}\approx \$12.29$$

.

68%- and 95%-Rule for the Sum of the Draws

Provided the number of draws is sufficiently large,

- for 68% of the time, the sum of the draws is within 1 SE from the expected value;
- for 95% of the time, the sum of the draws is within 2 SE from the expected value.

17.3 The Average of the Draws

Other than the sum of the draws, another quantity often of interest in a a box model is **the average of the draws**. The two are related as follows:

$$\text{Average of the draws} = \frac{\text{Sum of the draws}}{\text{Number of draws}}.$$

The expected value (E.V.) of the average of the draws is

$$\begin{aligned}\frac{\text{E.V. for the sum of draws}}{\text{Number of draws}} &= \frac{(\text{Number of draws}) \times (\text{Average of box})}{\text{Number of draws}} \\ &= \boxed{\text{Average of box}}.\end{aligned}$$

The standard error (SE) of the average of the draws is

$$\begin{aligned}\frac{\text{SE for the sum of draws}}{\text{Number of draws}} &= \frac{\sqrt{\text{Number of draws}} \times (\text{SD of box})}{\text{Number of draws}} \\ &= \boxed{\frac{\text{SD of box}}{\sqrt{\text{Number of draws}}}}.\end{aligned}$$

17.4 The Law of Large Number for a Box Model

The **law of large number** for a box model says that, as the number of draws goes up,

- the sum of the draws gets **farther away** from its expected value (number of draws) $\times$ (average of box),
- but the average of the draws gets **closer** to its expected value (average of box).

Back to The Law of Large Number in Chapter 16

What Does the Law of Averages Say?

Two fair coins are tossed 100 and 1000 times respectively.
Let N_{100} and N_{1000} be the respective numbers of heads.
Which of the following is more likely to be smaller?

$$|N_{100} - 50| \qquad\qquad |N_{1000} - 500|$$

Now let P_{100} and P_{1000} be be the percentage of heads[1].
Which of the following is more likely to be closer to 50%?

$$P_{100} \qquad\qquad P_{1000}$$

[1] Note $P_{100} = N_{100}/100 \times 100\%$, and $P_{1000} = N_{1000}/1000 \times 100\%$

Back to The Law of Large Number in Chapter 16

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Two fair coins are tossed 100 and 1000 times respectively. Let N_{100} and N_{1000} be the respective numbers of heads. Which of the following is more likely to be smaller?

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P_{100} $\qquad$ P_{1000}

[1] Note $P_{100} = N_{100}/100 \times 100\%$, and $P_{1000} = N_{1000}/1000 \times 100\%$

Back to The Law of Large Number in Chapter 16

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Which of the following is more likely to be smaller?

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$|N_{100} - 50|$ $\qquad$ $|N_{1000} - 500|$

Now let P_{100} and P_{1000} be be the percentage of heads[1].
Which of the following is more likely to be closer to 50%?

✓

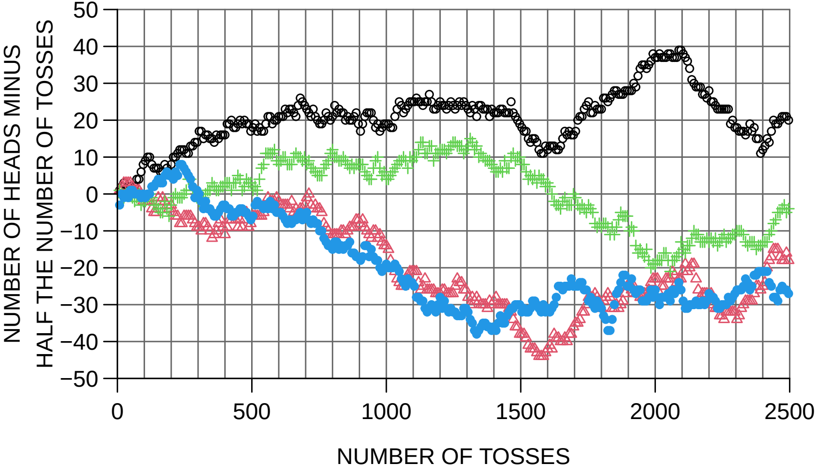
P_{100} $\qquad$ P_{1000}

[1] Note $P_{100} = N_{100}/100 \times 100\%$, and $P_{1000} = N_{1000}/1000 \times 100\%$

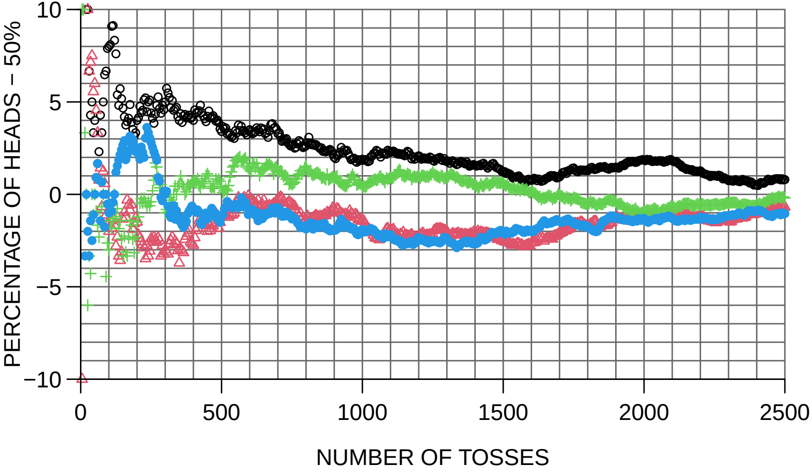
4 coins were tossed 2500 times each, and the "chance error"

$$\text{number of heads} - \text{half the number of tosses}$$

was plotted against the number of tosses:



For the same 4 coins, here are the chance errors expressed as a percentage of the number of tosses:



The Law of Averages

To summarize, what does the so-called **law of averages** have to say about the

$$\text{chance error} = \text{observed \# of heads} - \underbrace{\text{expected \# of heads}}_{\text{half the \# of tosses}}$$

*As the number of tosses goes up, the chance error gets **bigger** in absolute terms, but compared to the number of tosses, it gets **smaller**.*

A Box Model for Coin Tossing

The number of heads obtained in n tosses of a coin is like ____ draws without replacement from the box $\boxed{\boxed{0}\ \boxed{1}}$.

- Ave of Box =
- Expected number of H's (sum of draws) =
- SD of Box is

$$\left(\begin{array}{c}\text{bigger}\\\text{number}\end{array} - \begin{array}{c}\text{smaller}\\\text{number}\end{array}\right) \times \sqrt{\left(\begin{array}{c}\text{fraction with}\\\text{bigger number}\end{array}\right) \times \left(\begin{array}{c}\text{fraction with}\\\text{smaller number}\end{array}\right)}$$

$$= (1-0)\sqrt{\frac{1}{2} \times \frac{1}{2}} = 0.5$$

- SE of the number of H's = SE of the sum of draws equals

$$\sqrt{\text{Number of Draws}} \times (\text{SD of Box}) = \sqrt{n} \cdot 0.5$$

With more tosses, the diff

observed number of heads − expected number of heads

gets **bigger**

A Box Model for Coin Tossing

The number of heads obtained in n tosses of a coin is like $\underline{\ n\ }$ draws without replacement from the box $\boxed{\boxed{0}\ \boxed{1}}$.

- Ave of Box $=$
- Expected number of H's (sum of draws) $=$
- SD of Box is

$$\left(\begin{matrix}\text{bigger}\\ \text{number}\end{matrix} - \begin{matrix}\text{smaller}\\ \text{number}\end{matrix}\right) \times \sqrt{\begin{pmatrix}\text{fraction with}\\ \text{bigger number}\end{pmatrix} \times \begin{pmatrix}\text{fraction with}\\ \text{smaller number}\end{pmatrix}}$$

$$= (1-0)\sqrt{\frac{1}{2} \times \frac{1}{2}} = 0.5$$

- SE of the number of H's $=$ SE of the sum of draws equals

$$\sqrt{\text{Number of Draws}} \times (\text{SD of Box}) = \sqrt{n} \cdot 0.5$$

With more tosses, the diff

$$\text{observed number of heads} - \text{expected number of heads}$$

gets **bigger**

A Box Model for Coin Tossing

The number of heads obtained in n tosses of a coin is like $\underline{\ n\ }$ draws without replacement from the box $\boxed{\boxed{0}\ \boxed{1}}$.

- Ave of Box $= (0+1)/2 = 0.5$
- Expected number of H's (sum of draws) $= 0.5(\text{Number of Tosses})$
- SD of Box is

$$\begin{pmatrix}\text{bigger} \\ \text{number}\end{pmatrix} - \begin{pmatrix}\text{smaller} \\ \text{number}\end{pmatrix} \times \sqrt{\begin{pmatrix}\text{fraction with} \\ \text{bigger number}\end{pmatrix} \times \begin{pmatrix}\text{fraction with} \\ \text{smaller number}\end{pmatrix}}$$

$$= (1-0)\sqrt{\frac{1}{2} \times \frac{1}{2}} = 0.5$$

- SE of the number of H's = SE of the sum of draws equals

$$\sqrt{\text{Number of Draws}} \times (\text{SD of Box}) = \sqrt{n} \cdot 0.5$$

With more tosses, the diff

observed number of heads − expected number of heads

gets **bigger**

A Box Model for Coin Tossing (Continued)

However, in terms of percentages,

$$\text{the obs'd percentage of H's} = \frac{\text{the obs'd number of H's}}{\text{number of tosses}} \times 100\%$$

- The expected percentage of H's is 50%
- The SE for the percentage of H's is

$$\begin{aligned} \frac{\text{SE for number of H's}}{\text{number of tosses}} \times 100\% &= \frac{0.5\sqrt{\text{number of tosses}}}{\text{number of tosses}} \times 100\% \\ &= \frac{0.5}{\sqrt{\text{number of tosses}}} \times 100\%. \end{aligned}$$

In general, as the number of tosses goes up, the difference between the percentage of heads and 50% tends to get **smaller**.

In 100 tosses of a fair coin, we expect to get 50 heads;

- the SE for the number of heads obtained is

$$\sqrt{\text{number of tosses}} \times (\text{SD of Box}) = \sqrt{100} \times 0.5 = 5$$

- the number of heads obtained is within
  - 1 SE $= 5$ from expected, i.e., 50 ± 5, for 68% of the time;
  - 2 SE $= 10$ from expected, i.e., 50 ± 10, for 95% of the time;
- the percentage of heads is
  - between 45% and 55%, for 68% of the time,
  - between 40% and 60%, for 95% of the time.

In 10,000 tosses of a fair coin, we expect to get 5000 heads;

- the SE for the number of heads obtained is

$$\sqrt{\text{number of tosses}} \times (\text{SD of Box}) = \sqrt{10000} \times 0.5 = 50$$

- the number of heads obtained is within
  - 1 SE $= 50$ from expected, i.e., 5000 ± 50, for 68% of the time;
  - 2SE $= 100$ from expected, i.e., 5000 ± 100, for 95% of the time
- the percentage of heads is
  - between 49.5% and 50.5%, for 68% of the time,
  - between 49.0% and 51.0%, for 95% of the time.

Some Wrong Statements on the Law of Averages

- Suppose you are tossing a coin. If you get a lot of heads, then tails start coming up. Or if you get too many tails, the chance for heads goes up. In the long run, the number of heads and the number of tails even out.

- Suppose one tossed a coin 10,000 times, and got 5067 heads, i.e., got 67 more heads than the expected 5000 heads. As one tosses another 10,000 times, with 20000 tosses, the number of heads should be closer to the expected number.

Some Wrong Statements on the Law of Averages

- Suppose you are tossing a coin. If you get a lot of heads, then tails start coming up. Or if you get too many tails, the chance for heads goes up. In the long run, the number of heads and the number of tails even out.

- Suppose one tossed a coin 10,000 times, and got 5067 heads, i.e., got 67 more heads than the expected 5000 heads. As one tosses another 10,000 times, with 20000 tosses, the number of heads should be closer to the expected number.

Both statements are WRONG!

The law of averages does not work by compensation. When tossing a coin, for example, a run of heads is just as likely to be followed by a head as by a tail. A coin has no memory and no conscience.

Q: A coin will be tossed, and you will win \$1 if the number of heads is within 5 of half the number of tosses.
Which is better: 10 tosses or 100 tosses? Explain.
A: ______ tosses is better.

Q: A coin will be tossed, and you will win \$1 if the percentage of heads is between 40% and 60%.
Which is better: 10 tosses or 100 tosses? Explain.
A: _______ tosses is better.

Q: A coin will be tossed, and you will win \$1 if the number of heads is exactly equal to the number of tails.
Which is better: 10 tosses or 100 tosses? Explain.
A: ______ tosses is better.

Q: A coin will be tossed, and you will win \$1 if the number of heads is within 5 of half the number of tosses.
Which is better: 10 tosses or 100 tosses? Explain.
A: ___10___ tosses is better.

Q: A coin will be tossed, and you will win \$1 if the percentage of heads is between 40% and 60%.
Which is better: 10 tosses or 100 tosses? Explain.
A: _______ tosses is better.

Q: A coin will be tossed, and you will win \$1 if the number of heads is <u>exactly equal</u> to the number of tails.
Which is better: 10 tosses or 100 tosses? Explain.
A: ______ tosses is better.

Q: A coin will be tossed, and you will win \$1 if the number of heads is within 5 of half the number of tosses.
Which is better: 10 tosses or 100 tosses? Explain.
A: ___10___ tosses is better.

Q: A coin will be tossed, and you will win \$1 if the percentage of heads is between 40% and 60%.
Which is better: 10 tosses or 100 tosses? Explain.
A: ___100___ tosses is better.

Q: A coin will be tossed, and you will win \$1 if the number of heads is <u>exactly equal</u> to the number of tails.
Which is better: 10 tosses or 100 tosses? Explain.
A: ______ tosses is better.

Q: A coin will be tossed, and you will win \$1 if the number of heads is within 5 of half the number of tosses.
Which is better: 10 tosses or 100 tosses? Explain.
A: 10 tosses is better.

Q: A coin will be tossed, and you will win \$1 if the percentage of heads is between 40% and 60%.
Which is better: 10 tosses or 100 tosses? Explain.
A: 100 tosses is better.

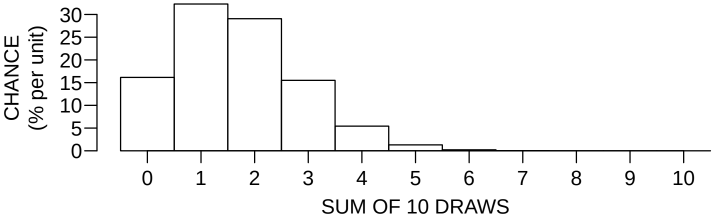
Q: A coin will be tossed, and you will win \$1 if the number of heads is exactly equal to the number of tails.
Which is better: 10 tosses or 100 tosses? Explain.
A: 10 tosses is better.

17.5 Probability Histogram

Consider the sum of 10 draws from the box $\boxed{1}\ \boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{0}$

- Ave. of box = 1/6, SD of box $= (1-0)\sqrt{(1/6)\times(5/6)} = \sqrt{5}/6$
- The sum of draws is expected to be ________________, give or take ___________________.

If we repeat the process "making 10 draws from the box and computing the sum of the draws" many many times, like 1,000,000 times, we can get 1,000,000 sums. What will the histogram of the 1,000,000 sums look like?



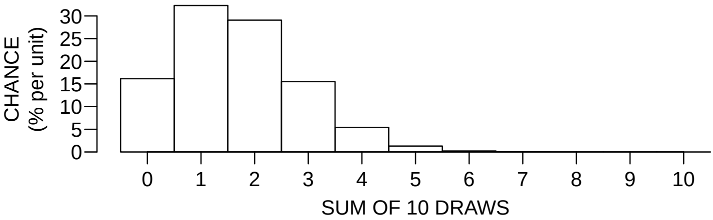
What does the area of the bar over 2 represent?

17.5 Probability Histogram

Consider the sum of 10 draws from the box $\boxed{1}\ \boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{0}$

- Ave. of box $= 1/6$, SD of box $= (1-0)\sqrt{(1/6)\times(5/6)} = \sqrt{5}/6$
- The sum of draws is expected to be $\underline{10 \times 1/6 \approx 1.67}$, give or take $\underline{\sqrt{10}\times\sqrt{5}/6 = 1.18}$.

If we repeat the process "making 10 draws from the box and computing the sum of the draws" many many times, like 1,000,000 times, we can get 1,000,000 sums. What will the histogram of the 1,000,000 sums look like?

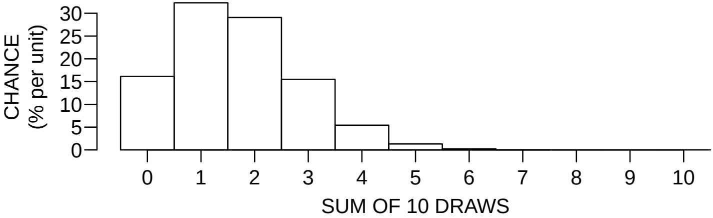


What does the area of the bar over 2 represent?

Probability Histogram (Cont'd)

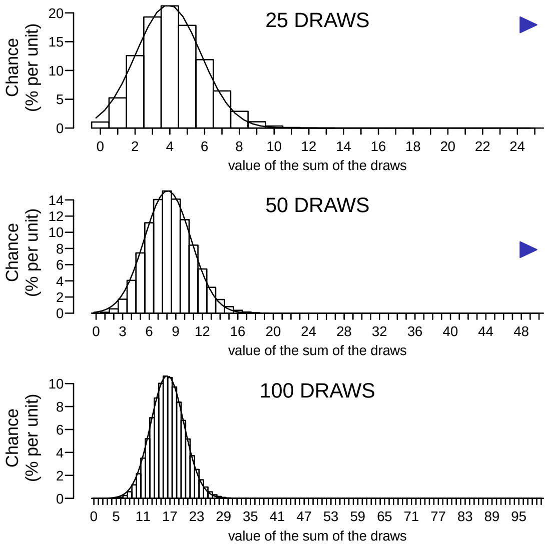
Probability histogram for the sum of 10 draws from the box

| 1 | 0 | 0 | 0 | 0 | 0 |



- A probability histogram represents **chance** by **area**.
- The total area of a probability histogram is **100%**.
- The area of the block over 2 represents the chance of get a sum of 2.
- The total area of the blocks over 0, 1, and 2 represents the chance of get a sum of 2 or less.

The figures below are the probability histograms for the sum of 25, 50, and 100 draws from the box $\boxed{\boxed{1}\ \boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{0}}$.

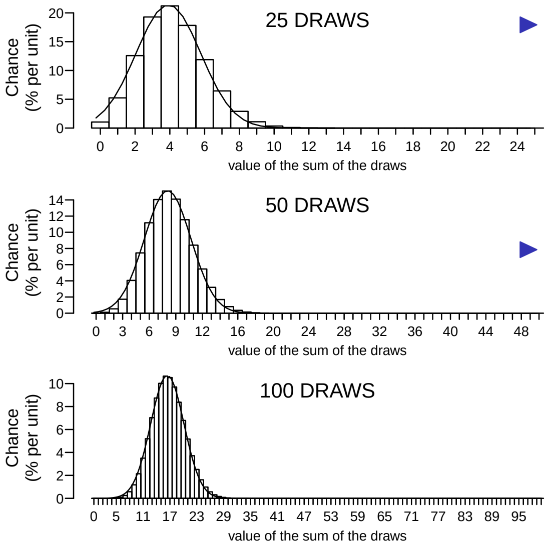


- As the number of draws goes up, the probability histogram follows the normal curve better and better
- For 100 draws, the chance that sum of the draws is between 10 and 20 draws can be approximated by the area under the

  ____

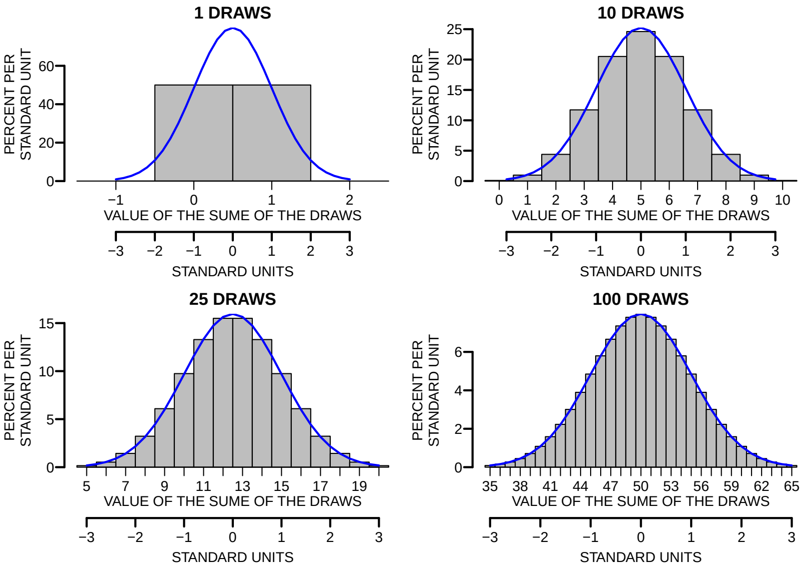
  between 10 and 20.

The figures below are the probability histograms for the sum of 25, 50, and 100 draws from the box $\boxed{1}\ \boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{0}\ \boxed{0}$.

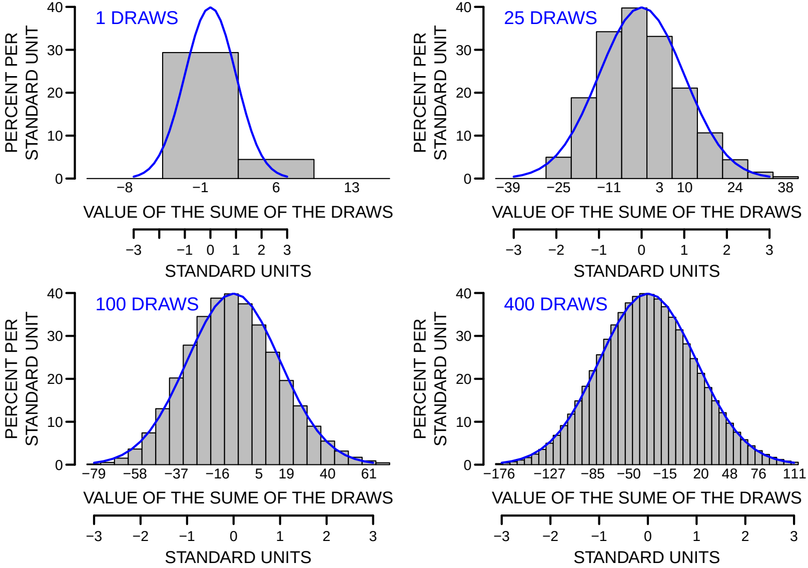


- As the number of draws goes up, the probability histogram follows the normal curve better and better
- For 100 draws, the chance that sum of the draws is between 10 and 20 draws can be approximated by the area under the <u>normal curve</u> between 10 and 20.

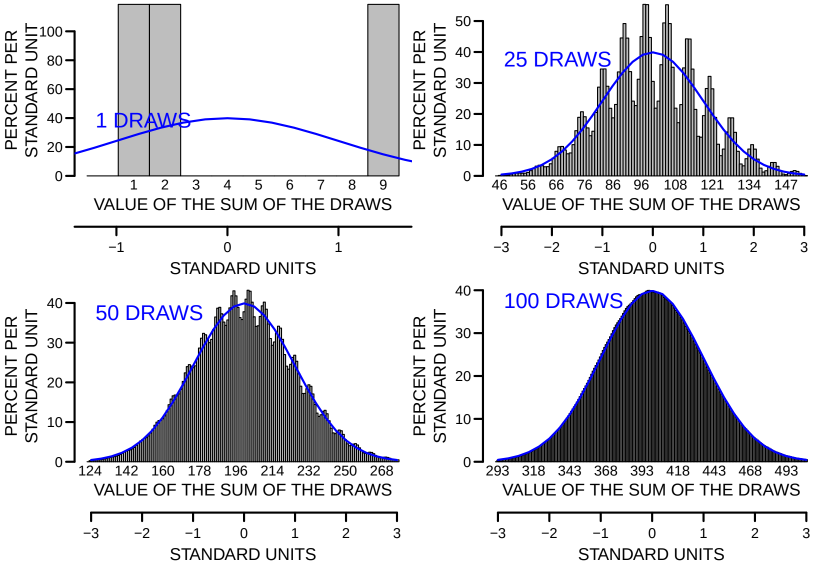
Probability histograms for the sum of 1, 10, 25 and 100 draws from the box $\boxed{\boxed{0}\ \boxed{1}}$.



Probability histograms for the sum of 1, 25, 100, and 400 draws from the box | 5 tickets of $\boxed{6}$, 33 tickets of $\boxed{-1}$ |.



Probability histograms for the sum of 1, 25, 50 and 100 draws from the box $\boxed{1}\ \boxed{2}\ \boxed{9}$.



Central Limit Theorem

Provided the number of draws is large enough, the probability histogram for the sum of the draws from **any box** follows the **normal curve**.

- With CLT, the **probability histogram** can be approximated by the **normal curve**, as long as the number of draws is large enough.
- The **center of the normal curve** is the **expected value of the sum**

$$(\text{number of draws}) \times (\text{average of the box}).$$

- The **SD of the normal curve** is **the standard error (SE) of the sum**

$$\sqrt{\text{number of draws}} \times (\text{SD of the box})$$

Using the Normal Curve I

A die is rolled 100 times, the total number of spots is expected to be _______ give or take _________. The chance that the sum of spots is 360 or over is ______

- The total number of spots in 100 rolls is like the sum of _______ draws from the box

$$\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}\ \boxed{6}$$

- Average of box = ________________________________
- SD of box =

- The expected value of the sum = __________________
- The SE of the sum = ________________________

Using the Normal Curve I

A die is rolled 100 times, the total number of spots is expected to be _______ give or take _________. The chance that the sum of spots is 360 or over is ______

- The total number of spots in 100 rolls is like the sum of __100__ draws from the box

$$\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}\ \boxed{6}$$

- Average of box = ________________________________
- SD of box =

- The expected value of the sum = __________________
- The SE of the sum = ________________________

Using the Normal Curve I

A die is rolled 100 times, the total number of spots is expected to be _______ give or take _________. The chance that the sum of spots is 360 or over is ______

- The total number of spots in 100 rolls is like the sum of __100__ draws from the box

$$\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}\ \boxed{6}$$

- Average of box = $(1 + 2 + 3 + 4 + 5 + 6)/6 = 3.5$
- SD of box =

- The expected value of the sum = ________________
- The SE of the sum = ________________________

Using the Normal Curve I

A die is rolled 100 times, the total number of spots is expected to be _______ give or take _________. The chance that the sum of spots is 360 or over is ______

- The total number of spots in 100 rolls is like the sum of __100__ draws from the box

$$\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}\ \boxed{6}$$

- Average of box = $(1+2+3+4+5+6)/6 = 3.5$
- SD of box =

$$\sqrt{\frac{(1-3.5)^2+(2-3.5)^2+\ldots+(6-3.5)^2}{6}} = 1.708.$$

- The expected value of the sum = __________________
- The SE of the sum = ________________________

Using the Normal Curve I

A die is rolled 100 times, the total number of spots is expected to be _______ give or take _________. The chance that the sum of spots is 360 or over is ______

- The total number of spots in 100 rolls is like the sum of __100__ draws from the box

$$\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}\ \boxed{6}$$

- Average of box = $(1+2+3+4+5+6)/6 = 3.5$
- SD of box =

$$\sqrt{\frac{(1-3.5)^2+(2-3.5)^2+\ldots+(6-3.5)^2}{6}} = 1.708.$$

- The expected value of the sum = $100 \times 3.5 = 350$
- The SE of the sum = ____________________

Using the Normal Curve I

A die is rolled 100 times, the total number of spots is expected to be _______ give or take _________. The chance that the sum of spots is 360 or over is ______

- The total number of spots in 100 rolls is like the sum of __100__ draws from the box

$$\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}\ \boxed{6}$$

- Average of box = $(1+2+3+4+5+6)/6 = 3.5$
- SD of box =

$$\sqrt{\frac{(1-3.5)^2+(2-3.5)^2+\ldots+(6-3.5)^2}{6}} = 1.708.$$

- The expected value of the sum = $100 \times 3.5 = 350$
- The SE of the sum = $\sqrt{100} \times 1.708 = 17.08$

Using the Normal Curve I

A die is rolled 100 times, the total number of spots is expected to be __350__ give or take _________. The chance that the sum of spots is 360 or over is ______

- The total number of spots in 100 rolls is like the sum of __100__ draws from the box

$$\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}\ \boxed{6}$$

- Average of box = $(1+2+3+4+5+6)/6 = 3.5$
- SD of box =

$$\sqrt{\frac{(1-3.5)^2 + (2-3.5)^2 + \ldots + (6-3.5)^2}{6}} = 1.708.$$

- The expected value of the sum = $100 \times 3.5 = 350$
- The SE of the sum = $\sqrt{100} \times 1.708 = 17.08$

Using the Normal Curve I

A die is rolled 100 times, the total number of spots is expected to be 350 give or take 17.08. The chance that the sum of spots is 360 or over is ______

- The total number of spots in 100 rolls is like the sum of 100 draws from the box

$$\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}\ \boxed{6}$$

- Average of box = $(1+2+3+4+5+6)/6 = 3.5$
- SD of box =

$$\sqrt{\frac{(1-3.5)^2+(2-3.5)^2+\ldots+(6-3.5)^2}{6}} = 1.708.$$

- The expected value of the sum = $100 \times 3.5 = 350$
- The SE of the sum = $\sqrt{100} \times 1.708 = 17.08$

Using the Normal Curve I

A die is rolled 100 times, the total number of spots is expected to be __350__ give or take __17.08__. The chance that the sum of spots is 360 or over is __??__

- The total number of spots in 100 rolls is like the sum of __100__ draws from the box

$$\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}\ \boxed{6}$$

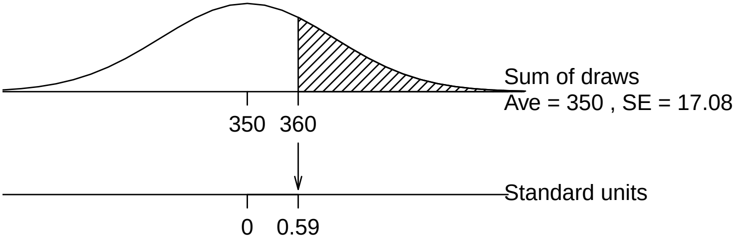
- Average of box = $(1+2+3+4+5+6)/6 = 3.5$
- SD of box =

$$\sqrt{\frac{(1-3.5)^2+(2-3.5)^2+\ldots+(6-3.5)^2}{6}} = 1.708.$$

- The expected value of the sum = $100 \times 3.5 = 350$
- The SE of the sum = $\sqrt{100} \times 1.708 = 17.08$

Using the Normal Curve I (Cont'd)

The chance that the sum of spots is 360 or over is approximately equal to the area under the normal curve to the ________ of 360.

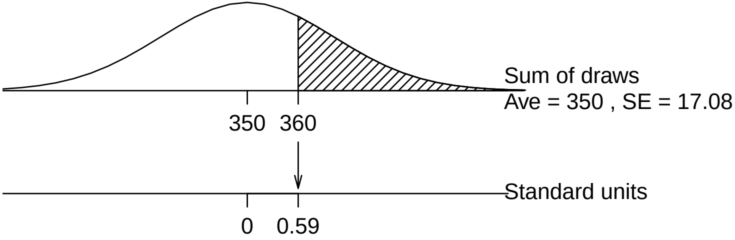


Standard units = ____________________

Chance ≈ ____________________________

Using the Normal Curve I (Cont'd)

The chance that the sum of spots is 360 or over is approximately equal to the area under the normal curve to the ___right___ of 360.

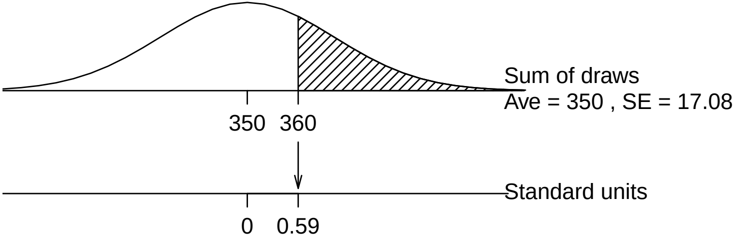


Standard units = ________________

Chance ≈ ______________________

Using the Normal Curve I (Cont'd)

The chance that the sum of spots is 360 or over is approximately equal to the area under the normal curve to the right of 360.

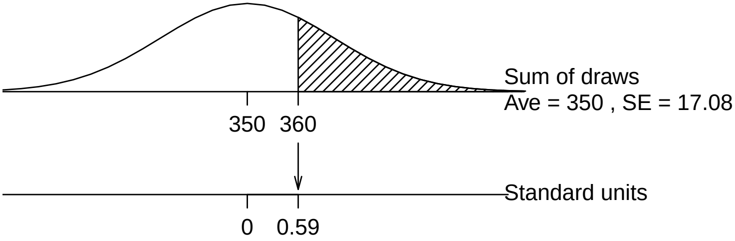


Standard units = $\dfrac{360 - 350}{17.08} \approx 0.59$

Chance ≈ ______________________

Using the Normal Curve I (Cont'd)

The chance that the sum of spots is 360 or over is approximately equal to the area under the normal curve to the right of 360.



Standard units $= \dfrac{360 - 350}{17.08} \approx 0.59$

Chance $\approx$ $1 - 0.7224 \approx 0.28 = 28\%$

Using the Normal Curve II

At Nevada roulette tables, the "house special" is a bet on the numbers 0, 00, 1, 2, and 3. The bet pays 6 to 1, and there are 5 chances in 38 to win. Someone plays roulette 100 times, betting a dollar on the house special each time. Estimate the chance that this person comes out ahead.

- The player's net gain is like the sum of _______draws at random with replacement from the box

$$\boxed{5 \text{ tickets of } \boxed{6} \qquad 33 \text{ tickets of } \boxed{-1}}$$

- The average of the box is

$$\frac{5 \times 6) + 33 \times (-1)}{38} \approx -0.08.$$

So the gambler expects to lose about _____cents per dollar bet.

Using the Normal Curve II

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- The player's net gain is like the sum of ____100____ draws at random with replacement from the box

$$\text{5 tickets of } \boxed{6} \qquad \text{33 tickets of } \boxed{-1}$$

- The average of the box is

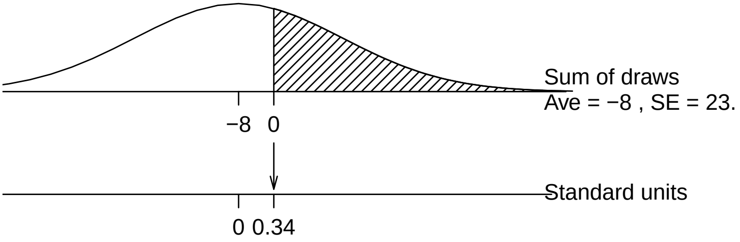
$$\frac{5 \times 6) + 33 \times (-1)}{38} \approx -0.08.$$

So the gambler expects to lose about ____8____ cents per dollar bet.

Using the Normal Curve II (Cont'd)

5 tickets of $\boxed{6}$	33 tickets of $\boxed{-1}$

- Average of box ≈ -0.08.
  SD of box $= (6 - (-1)) \times \sqrt{5/38 \times 33/38} \approx 2.37$.
- The player's net gain in 100 plays is expected to be
  , give or take .
- What is the chance that the player comes out ahead?



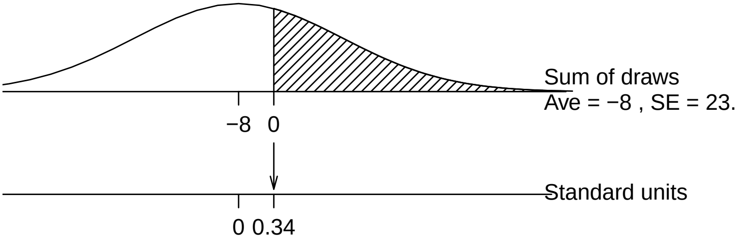
Standard units = ______________________

Chance ≈ __________________

Using the Normal Curve II (Cont'd)

5 tickets of $\boxed{6}$	33 tickets of $\boxed{-1}$

- Average of box ≈ -0.08.
  SD of box $= (6 - (-1)) \times \sqrt{5/38 \times 33/38} \approx 2.37$.
- The player's net gain in 100 plays is expected to be $100 \times -0.08 = -\$8$, give or take $\sqrt{100} \times 2.37 = \$23.7$.
- What is the chance that the player comes out ahead?



Standard units = ________________________

Chance $\approx$ ____________________

Using the Normal Curve II (Cont'd)

$\boxed{\text{5 tickets of } \boxed{6} \qquad \text{33 tickets of } \boxed{-1}}$

- Average of box ≈ -0.08.
  SD of box $= (6-(-1)) \times \sqrt{5/38 \times 33/38} \approx 2.37$.
- The player's net gain in 100 plays is expected to be $100 \times -0.08 = -\$8$, give or take $\sqrt{100} \times 2.37 = \$23.7$.
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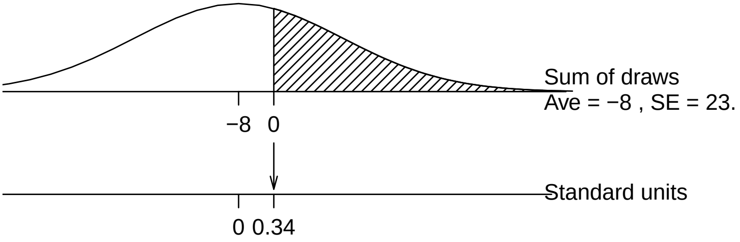
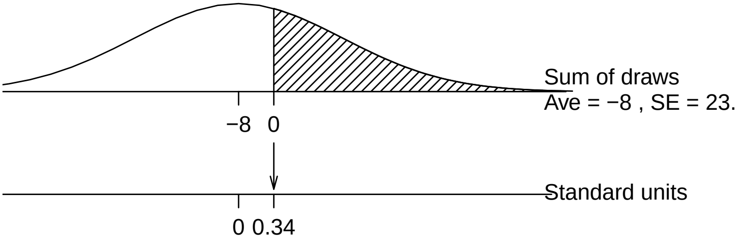


Standard units $=$ $\underline{(0-(-8))/23.7 \approx 0.34}$
Chance $\approx$ ________________

Using the Normal Curve II (Cont'd)

5 tickets of $\boxed{6}$ 33 tickets of $\boxed{-1}$

- Average of box ≈ -0.08.
  SD of box $= (6-(-1)) \times \sqrt{5/38 \times 33/38} \approx 2.37$.
- The player's net gain in 100 plays is expected to be $100 \times -0.08 = -\$8$, give or take $\sqrt{100} \times 2.37 = \$23.7$.
- What is the chance that the player comes out ahead?



Standard units = $(0-(-8))/23.7 \approx 0.34$

Chance $\approx$ 1-0.6331=36.69%

Summary for the Box Models

So far we have seen that when drawing at random with replacement from a box of numbered tickets:

- The expected value for the sum of the draws equals

$$(\text{number of draws}) \times (\text{average of box}).$$

  - This was "expected."
- The standard error for the sum of the draws equals

$$\sqrt{\text{number of draws}} \times (\text{SD of the box}).$$

  - This is remarkable.
- There is one more important fact — provided the number of draws is sufficiently large, the probability histogram for the sum of the draws will follow the normal curve, *even if the contents of the box do not*.
  - This is truly miraculous!
    But how large is "sufficiently large?' '

How Large the Number of Draws is Sufficiently Large?

If the tickets in the box ...

- follow the normal curve themselves, then any n will do;
- have a symmetric distribution, then n should be ≥ 25 or so;
- have a skewed or irregular distribution, then n should be moderate (≥ 100 or so), or even larger depending on how skewed or irregular the box is.

A Rule of Thumb: If the tickets in a box show **only two** different numbers, the number n of draws need to satisfy

$$n \times \begin{pmatrix} \text{fraction with} \\ \text{bigger number} \end{pmatrix} \geq 10 \quad \text{and} \quad n \times \begin{pmatrix} \text{fraction with} \\ \text{smaller number} \end{pmatrix} \geq 10.$$

Exercise

For each of the following 3 boxes, determine how large the number *n* of draws need to be to be using the rule of thumb.

- $\boxed{\boxed{0}\ \boxed{1}}$
- $\boxed{9\ \boxed{0}\text{'s}\quad \boxed{1}}$
- $\boxed{24\ \boxed{1}\text{'s}\quad \boxed{1}}$

Exercise

For each of the following 3 boxes, determine how large the number *n* of draws need to be to be using the rule of thumb.

- $\boxed{0}\ \boxed{1}$
- 9 $\boxed{0}$'s $\boxed{1}$
- 24 $\boxed{1}$'s $\boxed{1}$

The following are the probability histograms for the sums of 25 draws from each of the 3 boxes above

$\boxed{0}\ \boxed{1}$ | 9 $\boxed{0}$'s $\boxed{1}$ | 24 $\boxed{1}$'s $\boxed{1}$